SOC Security Investigation Report, Splunk

Case ID: SOC-2026-001
Investigation Type: Authentication Log Analysis
Platform: Splunk Enterprise
MITRE ATT&CK: T1110.003 - Password Spraying
Data Classification: Synthetic Training Data

1. Executive Summary

A security investigation was conducted in Splunk Enterprise to analyze authentication logs and identify patterns potentially associated with suspicious login activity.

The investigation identified a source IP, 203.0.113.77, associated with failed login attempts against 16 distinct accounts. This pattern is consistent with potential password spraying and warrants further review. The analysis also examined repeated account-level failures and successful logins following failed authentication events.

The available evidence does not confirm account compromise or establish malicious intent. The findings support further investigation and demonstrate a structured approach to authentication log analysis and SPL-based threat detection.

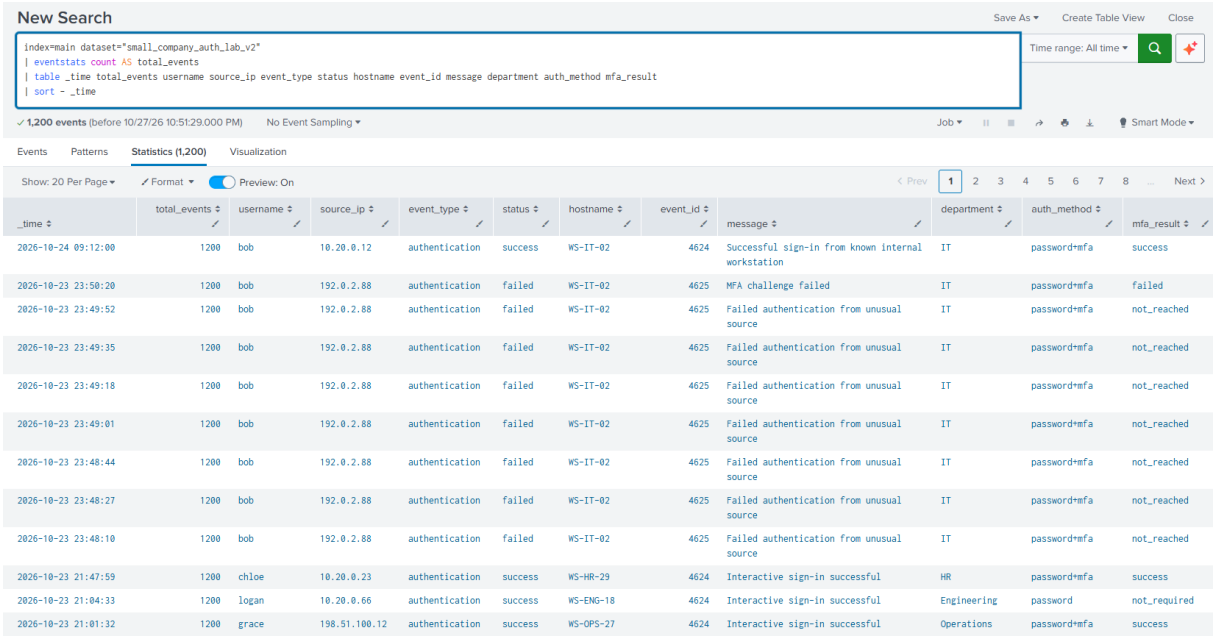
2. Investigation Methodology

The investigation used Splunk Search & Reporting to analyze events from the main index, filtered to the small_company_auth_lab_v2 dataset.

The analysis focused on:

- Authentication event review and relevant field inspection.
- Failed-login analysis by source IP and username.
- Identification of IP addresses targeting multiple accounts.
- Review of activity associated with a selected source IP.
- Comparison of failed and successful authentication events.
- Development of a threshold-based detection query.

Figure 1 - Authentication Event Overview



_time	total_events	username	source_ip	event_type	status	hostname	event_id	message	department	auth_method	mfa_result
2026-10-24 09:12:00	1200	bob	10.20.0.12	authentication	success	WS-IT-02	4624	Successful sign-in from known internal workstation	IT	password+mfa	success
2026-10-23 23:50:20	1200	bob	192.0.2.88	authentication	failed	WS-IT-02	4625	MFA challenge failed	IT	password+mfa	failed
2026-10-23 23:49:52	1200	bob	192.0.2.88	authentication	failed	WS-IT-02	4625	Failed authentication from unusual source	IT	password+mfa	not_reached
2026-10-23 23:49:35	1200	bob	192.0.2.88	authentication	failed	WS-IT-02	4625	Failed authentication from unusual source	IT	password+mfa	not_reached
2026-10-23 23:49:18	1200	bob	192.0.2.88	authentication	failed	WS-IT-02	4625	Failed authentication from unusual source	IT	password+mfa	not_reached
2026-10-23 23:49:01	1200	bob	192.0.2.88	authentication	failed	WS-IT-02	4625	Failed authentication from unusual source	IT	password+mfa	not_reached
2026-10-23 23:48:44	1200	bob	192.0.2.88	authentication	failed	WS-IT-02	4625	Failed authentication from unusual source	IT	password+mfa	not_reached
2026-10-23 23:48:27	1200	bob	192.0.2.88	authentication	failed	WS-IT-02	4625	Failed authentication from unusual source	IT	password+mfa	not_reached
2026-10-23 23:48:10	1200	bob	192.0.2.88	authentication	failed	WS-IT-02	4625	Failed authentication from unusual source	IT	password+mfa	not_reached
2026-10-23 21:47:59	1200	chloe	10.20.0.23	authentication	success	WS-HR-29	4624	Interactive sign-in successful	HR	password+mfa	success
2026-10-23 21:04:33	1200	logan	10.20.0.66	authentication	success	WS-ENG-18	4624	Interactive sign-in successful	Engineering	password	not_required
2026-10-23 21:01:32	1200	grace	198.51.100.12	authentication	success	WS-OPS-27	4624	Interactive sign-in successful	Operations	password+mfa	success

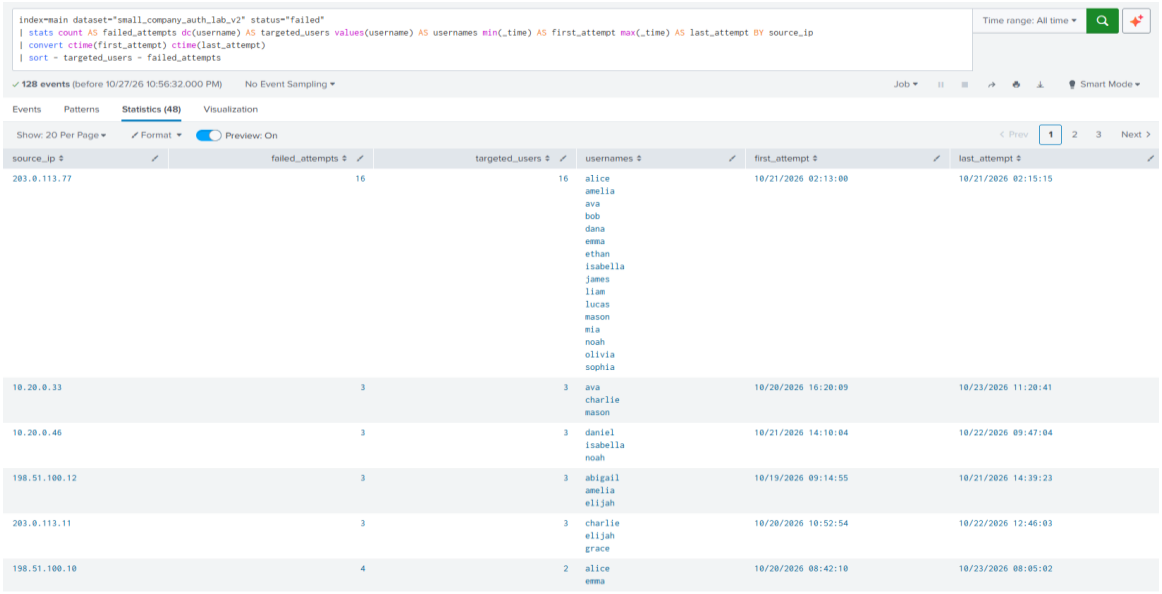
Authentication events reviewed in Splunk, including usernames, source IPs, outcomes, authentication methods, and MFA results.

3. Key Findings

3.1 Potential Password-Spraying Pattern

Source IP 203.0.113.77 was associated with failed authentication attempts involving **16 distinct accounts**. Repeated failures across multiple accounts can be consistent with password spraying, where an actor attempts credentials against several users.

Figure 2 - Failed Logins by Source IP



source_ip	failed_attempts	targeted_users	usernames	first_attempt	last_attempt
203.0.113.77	16	16	alice amelia ava bob dana emma ethan isabella james liam lucas mason mia noah olivia sophia	10/21/2026 02:13:00	10/21/2026 02:15:15
10.20.0.33	3	3	ava charlie mason	10/20/2026 16:20:09	10/23/2026 11:20:41
10.20.0.46	3	3	daniel isabella noah	10/21/2026 14:10:04	10/22/2026 09:47:04
198.51.100.12	3	3	abigail amelia elijah	10/19/2026 09:14:55	10/21/2026 14:39:23
203.0.113.11	3	3	charlie elijah grace	10/20/2026 10:52:54	10/22/2026 12:46:03
198.51.100.10	4	2	alice emma	10/20/2026 08:42:10	10/23/2026 08:05:02

Source IPs ranked by the number of targeted accounts and failed authentication attempts.

3.2 Repeated Account-Level Failures

The investigation also identified repeated failed logins involving the following accounts:

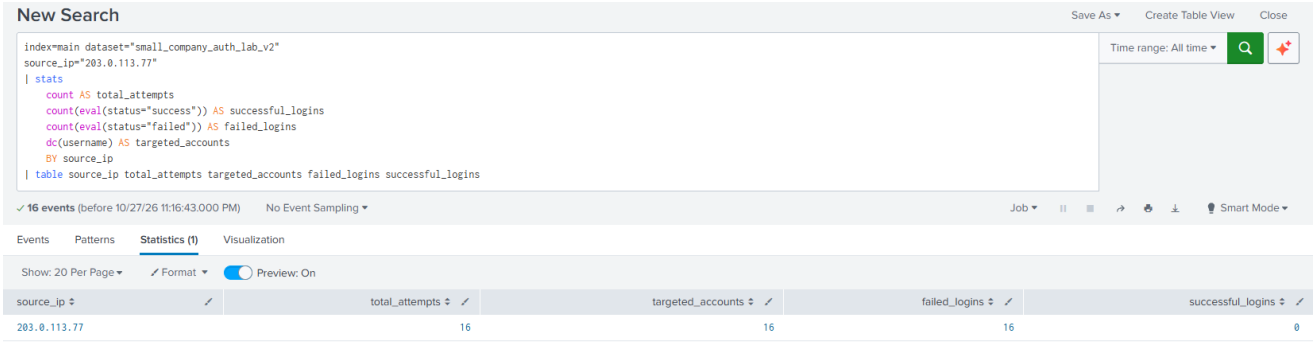
Account	Failed attempts
Charlie	18
Dana	14
Bob	8

These are separate account-level observations. They should not be attributed to 203.0.113.77 unless the corresponding source IP is confirmed in the event records.

3.3 Source IP Authentication Summary

A separate SPL query summarized authentication activity associated with 203.0.113.77, including the number of recorded events, distinct accounts, failed logins, and successful logins.

Figure 3 - Suspicious Source IP Summary



Summary of authentication activity associated with the selected source IP.

The failed-then-success query additionally examined successful events occurring within one hour of the immediately preceding failed event for the same username. Such matches require contextual review and do not independently prove that the successful login originated from the selected IP or resulted from an attack.

4. Risk Assessment and MITRE ATT&CK

Potential risk: Repeated authentication failures across multiple accounts may indicate password spraying or other suspicious authentication behavior.

MITRE ATT&CK mapping: T1110.003 - Password Spraying.

The observed pattern is consistent with this technique, but the mapping is an investigative hypothesis rather than confirmation of an actual attack.

5. Detection and Recommended Actions

A threshold-based SPL detection query was developed to identify source IPs generating failed authentication attempts against at least 10 distinct accounts within a 10 minute window.

Recommended defensive measures:

- Enforce multi-factor authentication (MFA), particularly for privileged and high risk accounts.
- Monitor repeated authentication failures across multiple accounts.
- Review successful logins following suspicious authentication activity.
- Tune detection thresholds to reduce false positives.
- Investigate matching events before taking account or network containment actions.

The detection logic is documented separately in detection-rule.spl. This documentation does not imply that a scheduled Splunk alert was configured or deployed.

6. Scope and Limitations

This investigation used synthetic authentication data in a Splunk Enterprise lab. The IP addresses belong to documentation ranges and do not identify real-world attackers.

The findings demonstrate authentication log analysis, SPL-based investigation, and detection logic development. No account compromise was confirmed, and no blocking or containment actions were performed.

Conclusion

The investigation identified authentication patterns warranting further review and produced a threshold-based query for potential password-spraying detection. The project demonstrates a structured SOC investigation workflow while clearly acknowledging the limitations of the available data.